



Lecture 3: **STYLE USING CSS**

3.1 Overviews of Cascading Style Sheet (CSS)

3.2 Configuring Color and Text with CSS

3.3 Visual Element and Graphics





3.1 Overviews of Cascading Style Sheet



What is CSS?

➤ stands for **Cascading Style Sheets**

➤ A language that describes the **style** of an HTML document

➤ **Styles** were added to HTML elements to be displayed

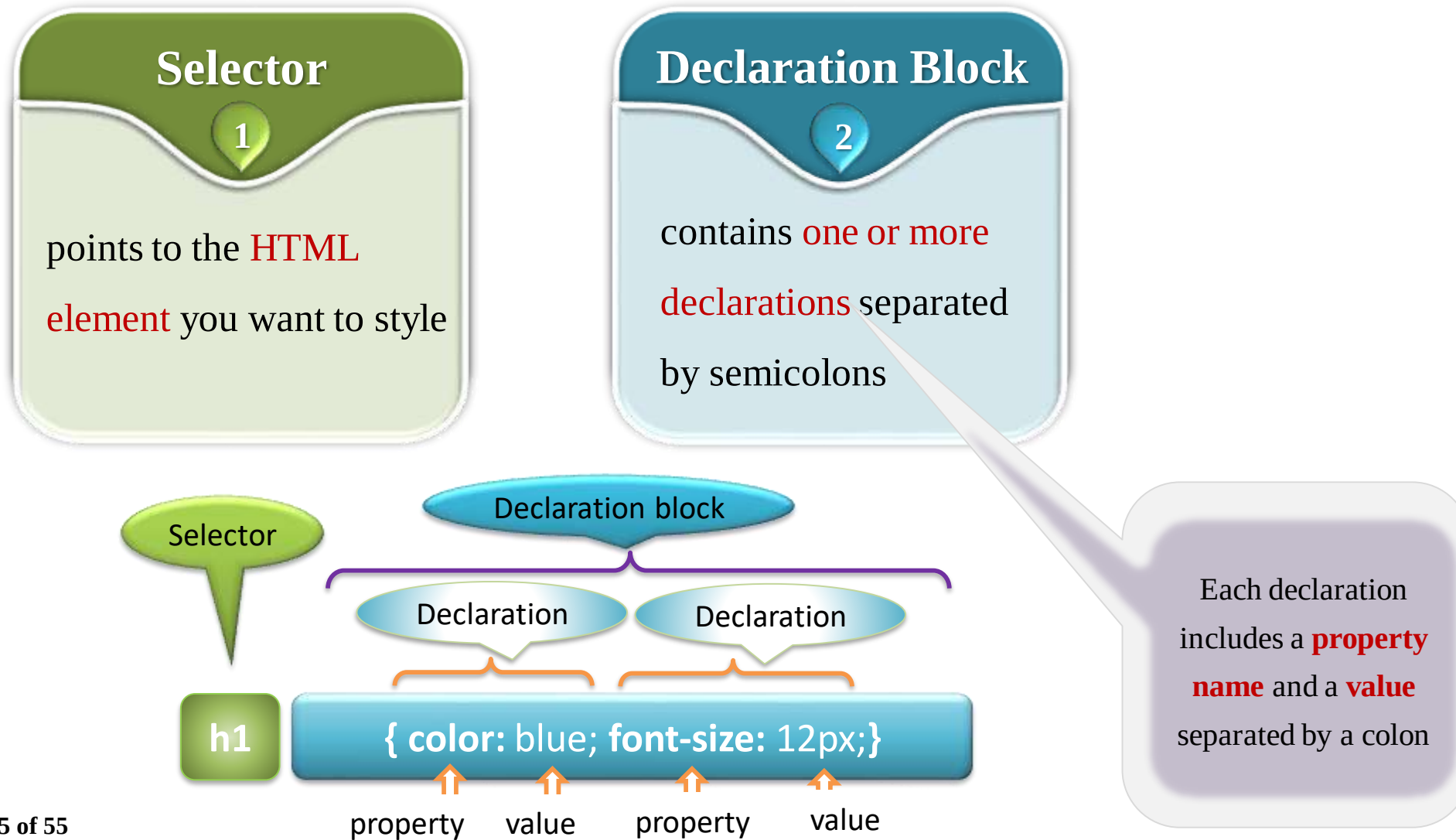
External Style Sheets are stored in CSS files



Advantages of Cascading Style Sheets

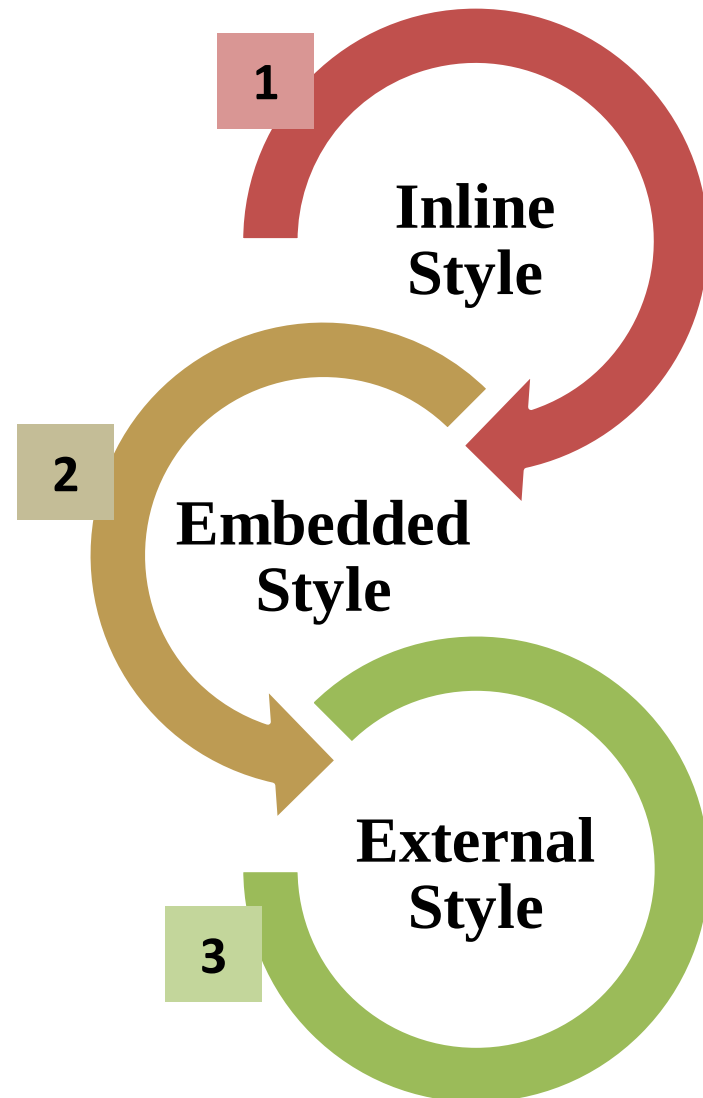
- 1 Typography and page layout can be better controlled.
- 2 Style is separate from structure.
- 3 Style can be stored.
- 4 It saves a lot of time.
- 5 Site maintenance is easier.

CSS Syntax





Configuring CSS





1. Inline CSS with the Style Attribute

- An inline style may be used to apply a **unique style** for an **individual element**.
- To use inline styles, add the **style attribute** to the relevant tag.
 - ✓ The style attribute can contain **any CSS property**.
- The example shows **how to change the color and the left margin** of a **<h1>** element:

```
<h1 style="color: blue; margin-left: 30px;">This is a heading.</h1>
```

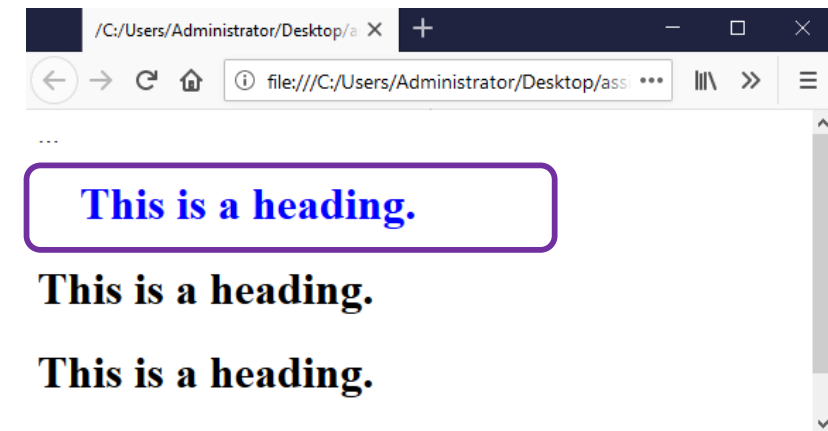



1. Web Page using Inline Style

Inline
style

```
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File Edit Format View Help

<!DOCTYPE html>
<html>
<head>...</head>
<body>
  <h1 style="color: blue; margin-left: 30px;">
    This is a heading.</h1>
  <h1>This is a heading.</h1>
  <h1>This is a heading.</h1>
</body>
</html>
```



2.

Embedded CSS with the Style Attribute

Embed
ded
style

➤ defines in
the **header** of
a web page

➤ apply style
instructions to
the **entire web
page**
document

➤ opening
<style> tag
and closing
</style> tag
contain the list
of embedded
style rules

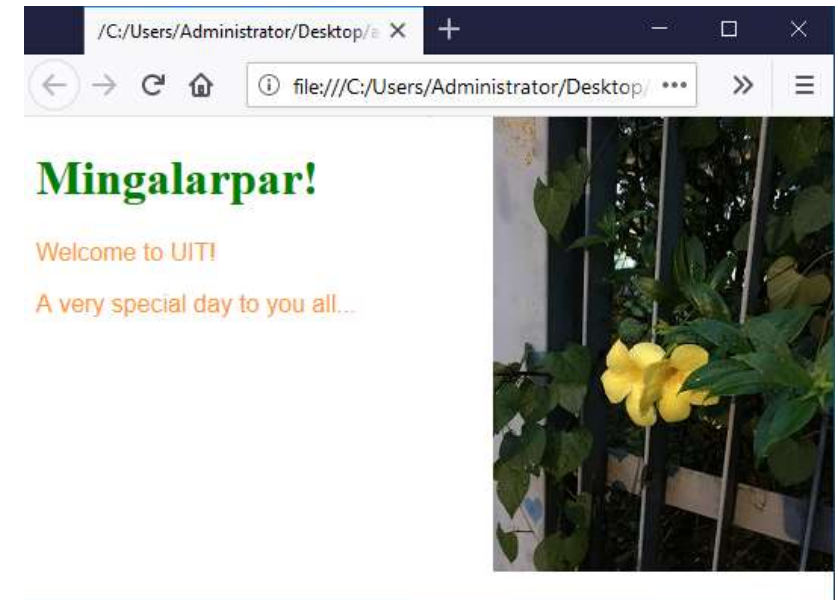


2.

Web Page using Embedded Style

Embedd
ed
style

```
<!DOCTYPE html>
<html>
<head>
  <style>
    body {
      background-image: url("image.jpg");
      background-repeat: no-repeat;
      background-position: right top;
      margin-right: 20px;}
    h1 {color: green; text-align: left;}
    p {color:#ff9933; text-align: left; font-family:
      Arial;}
  </style>
</head>
<body>
  <h1>Mingalarpar!</h1>
  <p>welcome to UIT!</p> <p>A very special day to you
  all..</p>
</body>
```



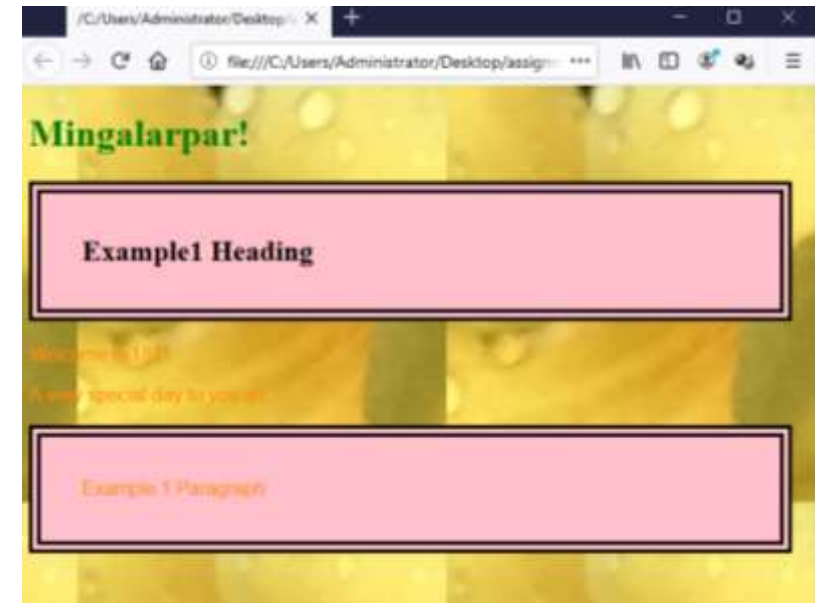


2.

Web Page using Embedded Style

Embedd
ed
style

```
<!DOCTYPE html>
<html>
<head>
  <style>
    body {
      background-image: url("image.jpg"); background-
      repeat: no-repeat; background-position: right
      top;
      margin-right: 20px;}
    h1 {color: green; text-align: left;}
    p {color:#ff8c00; text-align: left; font-family:
      Arial;}
    #example1 { border: 10px double black; padding:35px;
      background: pink; background-clip: border-box; }
  </style>
</head>
<body>
  <h1>Mingalarpar!</h1> <h2 id= example1> Example1
  Heading</h2>
  <p>welcome to UIT!</p> <p>A very special day to you
  all..</p>
```





3.

External Style Sheet

External
style

- With an external style sheet, you can change the look of an entire website by changing just one file!
- Each page must include a reference to the external style sheet file inside the **<link>** element.
 - The **<link>** element goes inside the head section:

```
<head>
```

```
<link rel="stylesheet" type="text/css"
```

```
href="mystyle.css">
```

```
</head>
```

".html" file



3.

External Style Sheet (cont'd)

⑩ An external style sheet **can be written in any text editor.**

⑩ The file **should not contain any html tags.**

⑩ The style sheet file must be saved with a **.css** extension.

```
body {
    background-color: lightblue;}
h1 {
    color: navy;
    margin-left: 20px;}
```

myStyle.css



CSS Properties

Property	Description	Possible Values	Examples
background-image	sets the background image for an element	(one of more) url	body{url("tree.png"), url("paper.gif"); }
background-attachment	specifies whether the background image will scroll with the page content or in a fixed position).	fixed scroll	body { background-attachment: fixed; }
background-color	Declares the background color .	Valid color names RGB values hexadecimal notation	body { background-color: red; } body { background-color: rgb(255,0,0); } body {background-color: #00FF00; }
font-family	Declares the name of the font to be used. Previously set in HTML via the face attribute in a tag.	Arial, Verdana, Sans-serif, Times New Roman, etc.	p{ font-family: Arial; } p{ font-family: "Arial", Helvetica, sans-serif; }
font-size	Declares the size of the font . Previously set in HTML via the size attribute in a tag.	1em, 12pt, 10px, 80% xx-small, x-small, small, medium, large, x-large, xx-large, smaller, larger	p {font-size: 70%; } p {font-size: 10px; } p {font-size: medium; }

¹⁴ Note: If the name of a font family is more than one word, it must be in quotation marks, like: "Arial".



CSS Properties (cont'd)

Property	Description	Possible Values	Examples
font-style	Declares the font style.	normal, italic, oblique	p { font-style: italic; } p { font-style: oblique; }
font-weight	Declares the font weight (lightness or boldness)	normal, bold, bolder, lighter, 100, 200	p { font-weight: bold; } p { font-weight: 200; }
margin	Declares the margin for the element.	{margin- top margin- right margin- bottom margin- left } with lengths, percentages, and auto which means the browser calculates.	p { margin: 2px 50px 75px 30px; } p { margin: 2px 50px 75px; } p { margin: 25px 50px; }
line-height	Declares the distance between lines of the element.	numbers, percentages, lengths, and the predefined value of normal .	p { line-height: normal; } p { line-height: 2em; } p { line-height: 125%; }

Note:

If the margin property has three values: margin: 25px 50px 75px; then top margin is 25px, right and left margins are 50px, bottom margin is 75px

If the margin property has two values then top and bottom margins are 25px, right and left margins are 50px

CSS Properties (cont'd)

Property	Description	Possible Values	Examples
text-align	Declares the horizontal alignment (left, right, center, justify) of inline content.	• left • right • center • justify	<pre>p { text-align: center; } p { text-align: right; }</pre>
text-decoration	Declares the text decoration.	• none • underline • overline • line-through • blink	<pre>div { text-decoration: none; } div { text-decoration: underline; }</pre>
width	Declares the width of the element .	• Lengths, • percentages	<pre>div { width: 500px; } div { width: 75%; }</pre>
color	Declares the color of the text .	• Valid color names, • RGB values, • hexadecimal notation.	<pre>div { color: green; } div { color: rgb(0,255,0); } div { color: #00FF00; }</pre>
text-indent	Declares the indentation of the first line of text .	• Lengths • percentages	<pre>div { text-indent: 12px; } div { text-indent: 2%; }</pre>



Summary

Lecture 3.1: Overviews of Cascading Style Sheet

- ✓ What is CSS?
- ✓ Advantages of CSS
- ✓ CSS Syntax
- ✓ Configuring CSS
 - Inline CSS
 - Embedded CSS
 - External CSS
- ✓ CSS Properties



3.2 Configuring —— Color and Text —— with CSS



WEB DESIGN

- Three types of website organization:

1.

Hierarchical

2.

Linear

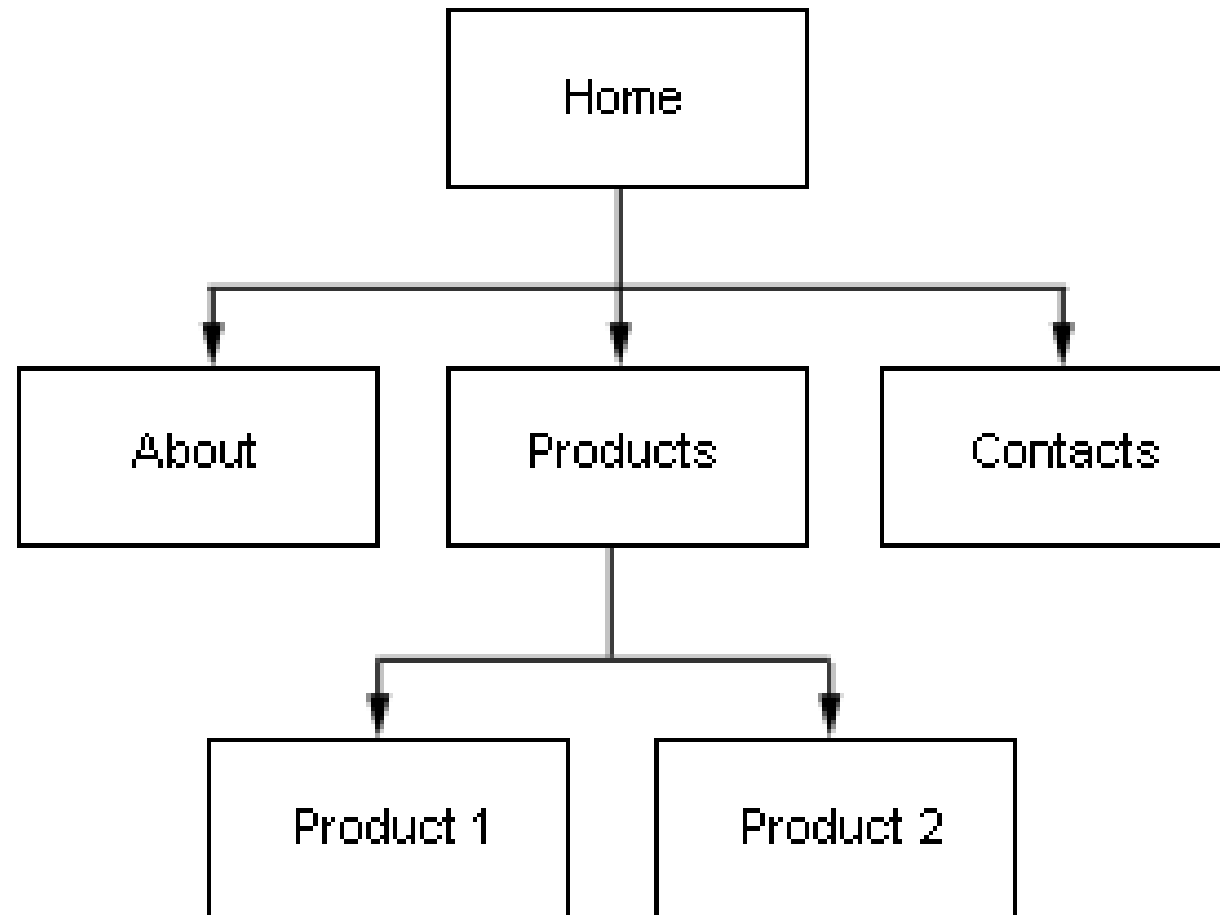
3.

Random (sometimes called Web organization)



1. Hierarchical Organization

hierarch
ical





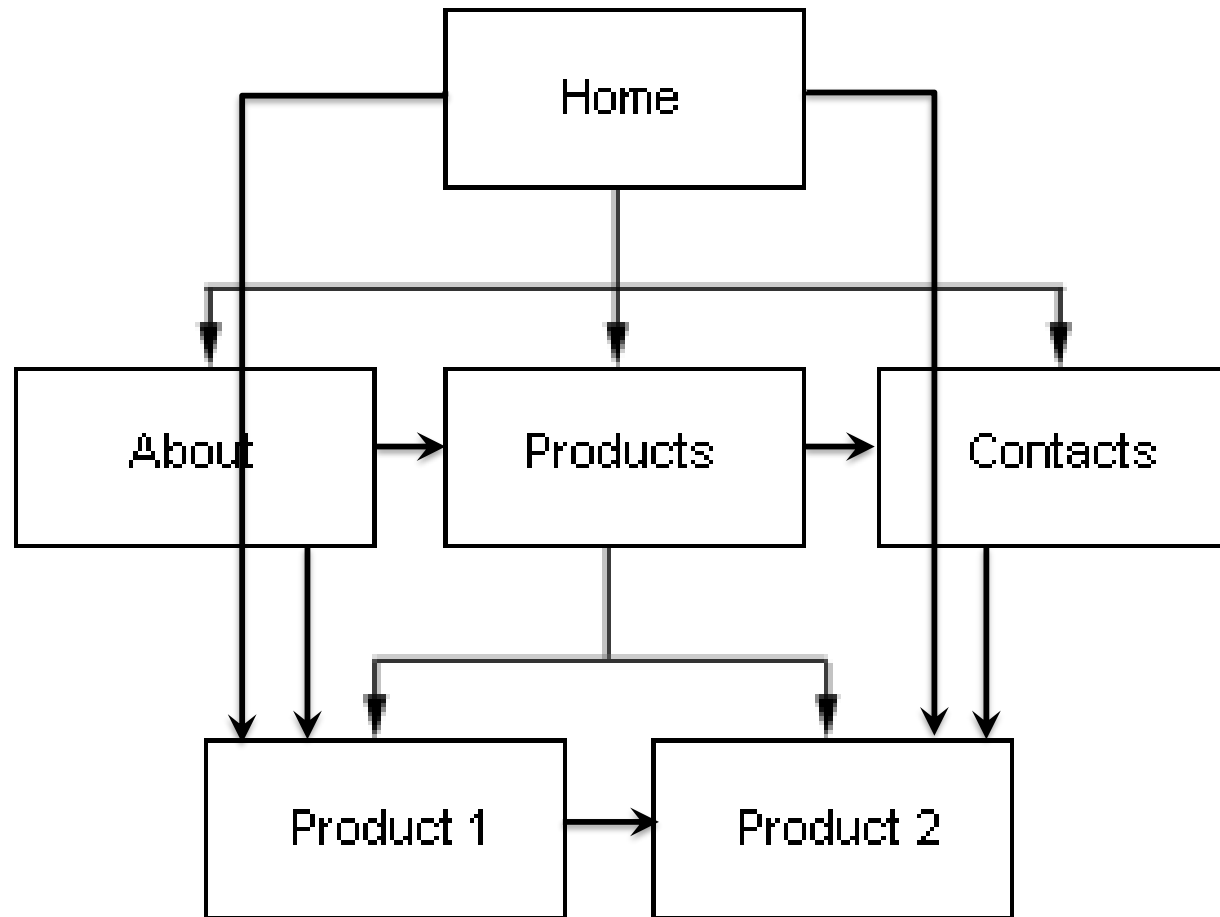
2. Linear Organization

linear



3. Random Organization

random





Span Element



defines an **inline container** for *elements* for *styling purposes*.



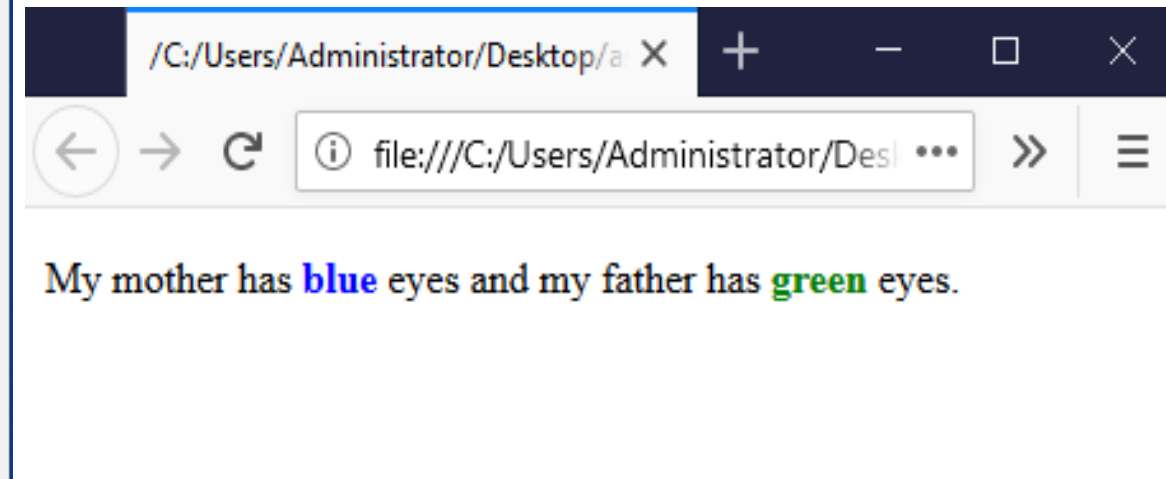
used to **style parts of an element**.



**** tag: Example

```
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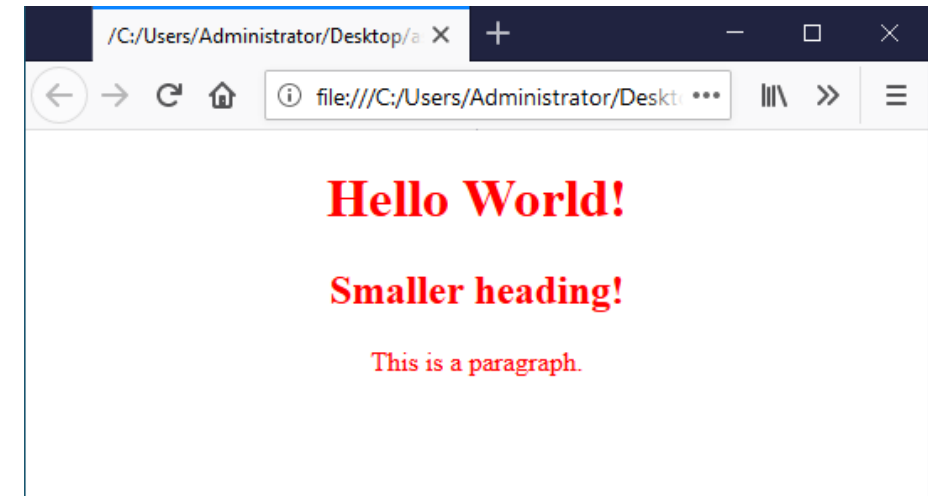
<!DOCTYPE html>
<html>
<body>
<p>My mother has <span style="color:
blue; font-weight:bold">blue</span>
eyes and my father has <span
style="color: green; font-weight:
bold">green</span> eyes.</p>
</body>
</html>
```





Grouping Selector

```
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<!DOCTYPE html>
<html>
<head>
  <style>
    h1, h2, p {
      text-align: center;
      color: red;
    }
  </style>
</head>
<body>
  <h1> Hello World!</h1>
  <h2> Smaller heading!</h2>
  <p> This is a paragraph.</p>
</body>
</html>
```





Properties in One Declaration

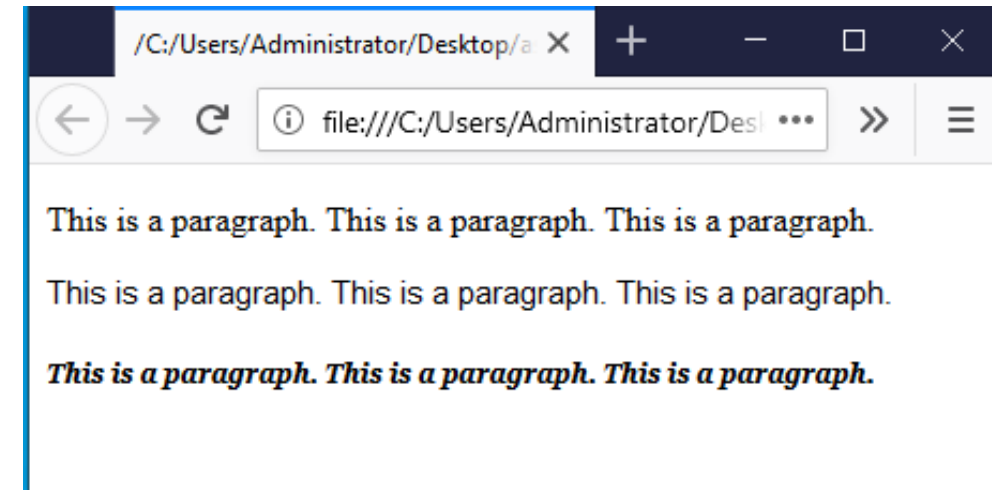
Untitled - Notepad

File Edit Format View Help

```

<!DOCTYPE html>
<html>
  <head>
    <style>
      p.ex1 {
        font: 15px arial, sans-serif; }
      p.ex2 {
        font: italic bold 12px/30px Georgia, serif;
      }
    </style>
  </head>
  <body>
    <p> This is a paragraph. This is a paragraph. This is a
      paragraph. </p>
    <p class="ex1"> This is a paragraph. This is a
      paragraph. This is a paragraph. </p>
    <p class="ex2"> This is a paragraph. This is a
      paragraph. This is a paragraph.</p>
  </body>
</html>
  
```

26 of 53



The **border** Property

- sets all the border properties *in one declaration*.
- The properties that can be set, are *(in order)*:



Example:

```
p { border: 5px solid red; }
```

- It does not matter if one of the values above are missing.



The border-style Property

- The type of line displayed in the border is configured.
- The formatting options include *inset*, *outset*, *double*, *groove*, *ridge*, *solid*, *dashed*, and *dotted*. E.g.,

```
h1{ border-width: 3px;  
    border-style: dashed;  
    border-color: #000033; }
```

- To shorten the code, it is also possible to specify all the individual border properties in one property.

E.g.,

```
h2{ border: 3px dashed #000033; }
```

border-width

border-style

border-color

border-style: Example

```
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File Edit Format View Help
<!DOCTYPE html>
<html>
<head>
<style>
h1 { border: 5px solid red; }
h2 { border: 4px dotted purple; }
div { border: double; }
</style>
</head>
<body>
<h1>A heading with a solid green border</h1>
<h2>A heading with a dotted purple border</h2>
<div>A div element with a double border.</div>
</body>
</html>
```





The Border Properties

Property		Description
border	border : solid; border : dashed red;	Configure the values for border-width, border-style, and border-color of an element
border-bottom	border-bottom : solid;	Configure the bottom border of an element
border-bottom-left-radius	border-bottom-left-radius : 50%;	Define the shape of the border of the bottom-left corner
border-bottom-right-radius	border-bottom-right-radius : 20px;	Define the shape of the border of the bottom-right corner
border-color	border-color : coral;	Specify the color of the border. Default value is the color of the element
border-left	border-left : solid;	Configure the left border of an element



The Border Properties (cont'd)

Property		Description
border-style	border-style : dotted;	Specify the style of the border. Default value is "none"
border-top	border-top : dashed red;	Configure the top border of an element
border-top-left-radius	border-top-left-radius : 50%;	Configure a rounded top left corner. This CSS3 property is not supported in all browsers.
border-top-right-radius	border-top-right-radius : 50%;	Configure a rounded top right corner. This CSS3 property is not supported in all browsers.
border-width	border-width : thick;	Specify the width of the border. Default value is "medium"



The **padding** Property

- generate **empty space** around the **content** of the HTML element and the **border**
- padding is **0 by default**
- shorthand property sets **all the padding properties** in one declaration
- property can have from **one to four values**

Example

Set the padding of a **<p> element**:

```
p { padding: 2cm 4cm 3cm 4cm; }
```

top right bottom left

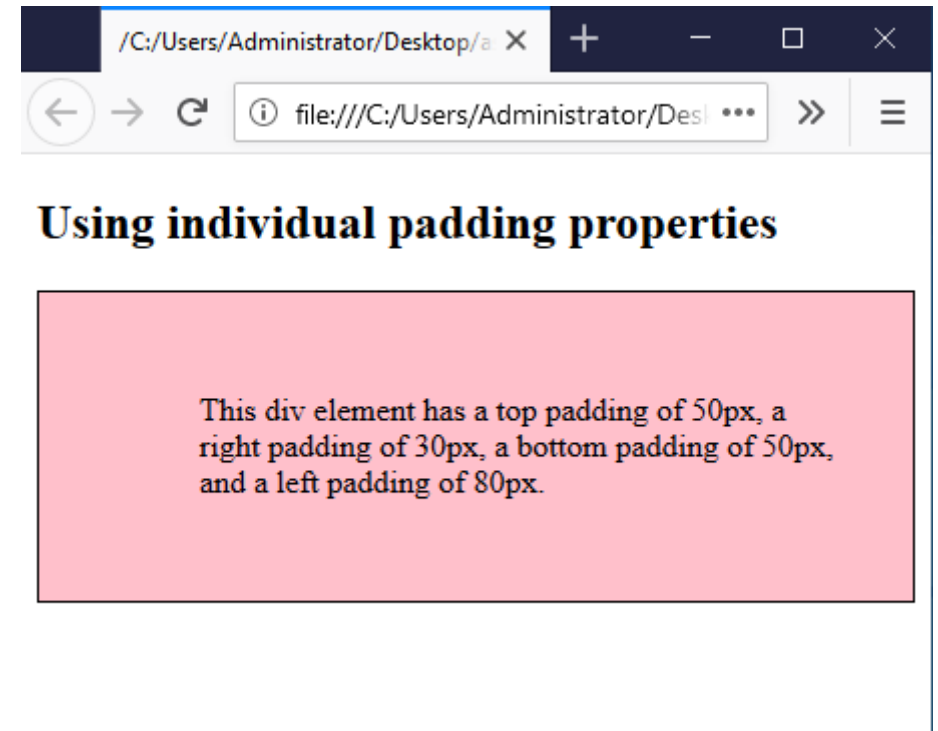




Padding: Example

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
<head> <style>
div {
  border: 1px solid black; background-color: pink;
  padding-top: 50px; padding-right: 30px;
  padding-bottom: 50px; padding-left: 80px; }
</style> </head>
<body> <h2>Using individual padding properties</h2>
<div>This div element has a top padding of 50px, a right
padding of 30px, a bottom padding of 50px, and a left
padding of 80px.</div>
</body>
</html>
```





The padding Property

Property	Description
padding	define the space between the element border and the element content
padding-bottom	blank space between an element and its bottom border
padding-left	blank space between an element and its left border
padding-right	blank space between an element and its right border
padding-top	blank space between an element and its top border



Lets practice!

Write a simple web page including all of the following codes

1. Add span tag,
2. Add grouping sector,
3. Add different border properites
4. Change to padding-right





Summary

Lecture 3.2: Configuring Color and Text with CSS

✓ WEB DESIGN

- Hierarchical Organization
- Linear Organization
- Random Organization

✓ Span Element

✓ Grouping Selector

✓ All the font properties in One Declaration

✓ The border and padding Property

3.3 Visual Element and Graphics





CSS3 background-clip Property

- The CSS3 background-clip property **specifies the painting area** of the background.
- The property takes three different values:
 1. **border-box** - (default) the background is painted to the **outside edge of the border**
 2. **padding-box** - the background is painted to the **outside edge of the padding**
 3. **content-box** - the background is **painted within the content box**



Background clip Example

```
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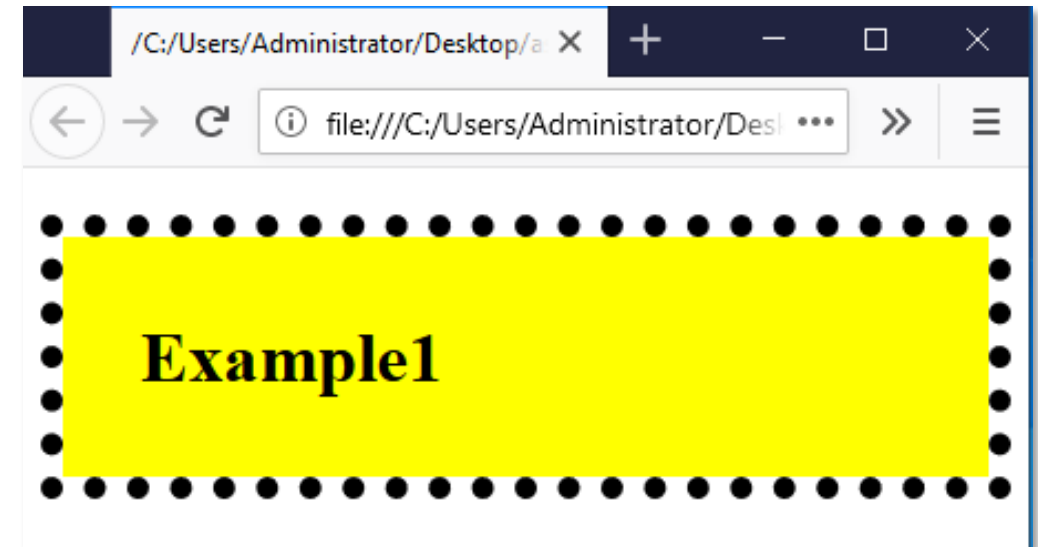
<!DOCTYPE html>
<html>
  <head>
    <style>
      #example1 { border: 10px dotted black;
                  padding: 35px; background: yellow;
                  background-clip: border-box; }
    </style>
  </head>
  <body>
    <h1 id= example1> Example1</h1>
  </body>
</html>
```



Background clip Example

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
  <head>
    <style>
      #example1 { border: 10px dotted black;
                  padding: 35px; background: yellow;
                  background-clip: padding-box; }
    </style>
  </head>
  <body>
    <h1 id= example1> Example1</h1>
  </body>
</html>
```

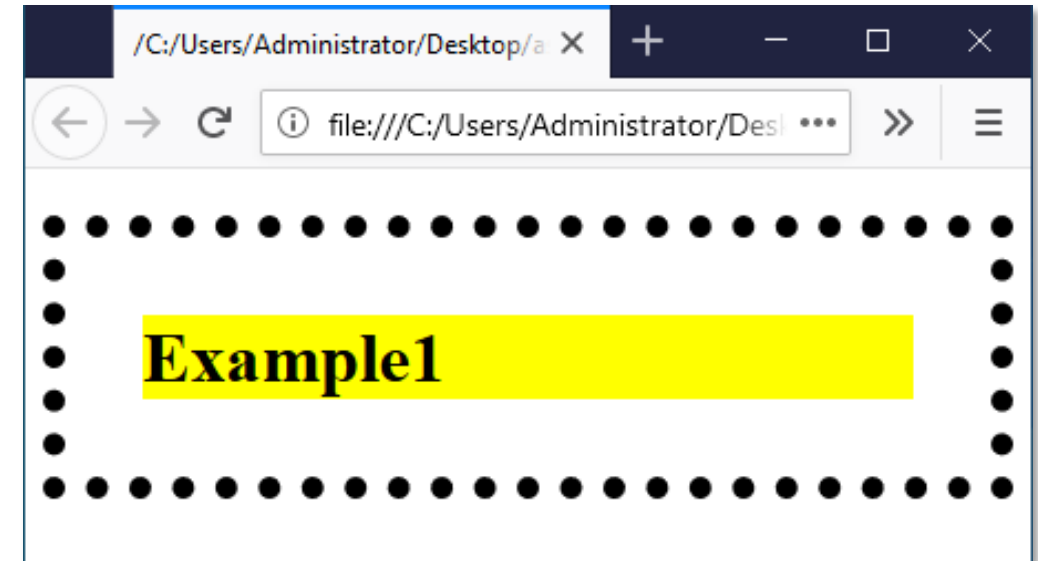




Background clip Example

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
  <head>
    <style>
      #example1 { border: 10px dotted black;
        padding: 35px; background: yellow;
        background-clip: content-box; }
    </style>
  </head>
  <body>
    <h1 id= example1> Example1</h1>
  </body>
</html>
```





CSS3 background-origin Property

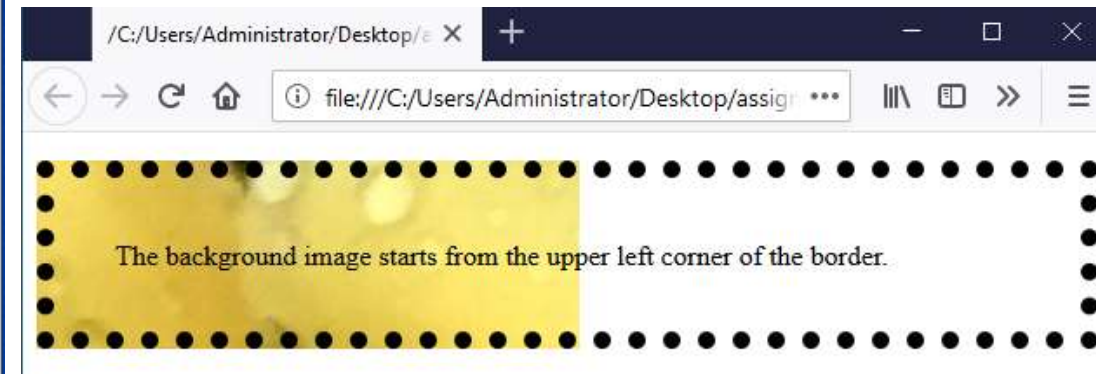
- The CSS3 background-origin property specifies **the origin position** (the background positioning area) of a background image.
- The property takes three different values:
 1. **border-box** - the background image starts from the **upper left corner of the border**
 2. **padding-box** - (default) the background image starts from the **upper left corner of the padding edge**
 3. **content-box** - the background image starts from the **upper left corner of the content**



Background origin Example

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
  <head>
    <style>
      #example1 {border: 10px dotted black;
padding:35px; background: url(image.jpg);
background-repeat: no-repeat; background-
origin: border-box;}
    </style>
  </head>
  <body> <p id= example1> The background image
starts from the upper left corner of the
border.</p>
  </body>
</html>
```

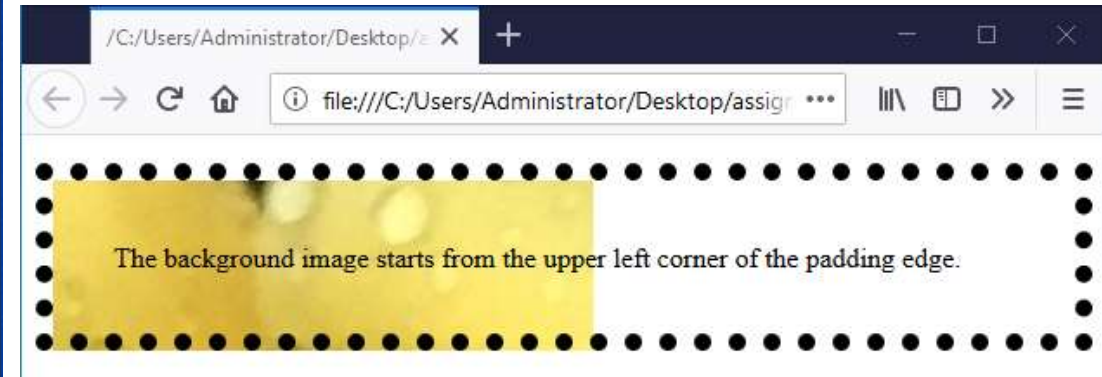




Background origin Example

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
  <head>
    <style>
      #example1 {border: 10px dotted black;
padding: 35px; background: url(image.jpg);
background-repeat: no-repeat; background-
origin: padding-box; }
    </style>
  </head>
  <body> <p id= example1> The background image
starts from the upper left corner of the
padding edge.</p>
  </body>
</html>
```

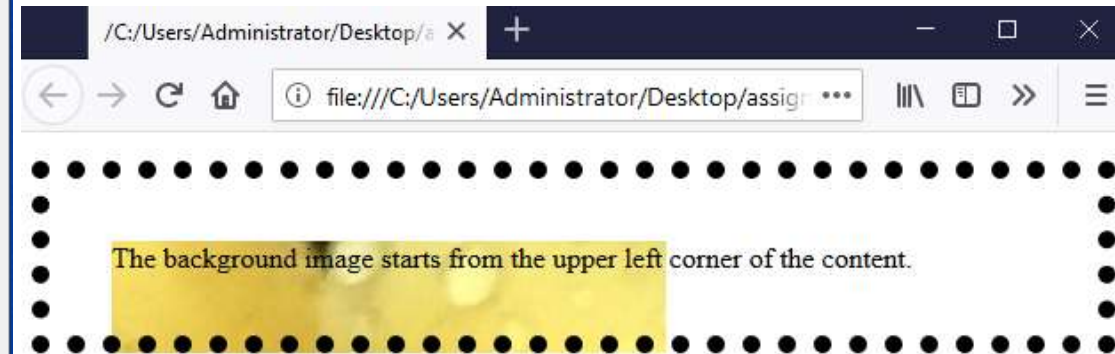




Background origin Example

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
  <head>
    <style>
      #example1 {border: 10px dotted black;
padding: 35px; background: url(image.jpg);
background-repeat: no-repeat; background-
origin: content-box; }
    </style>
  </head>
  <body> <p id= example1> The background image
starts from the upper left corner of the
content.</p>
  </body>
</html>
```



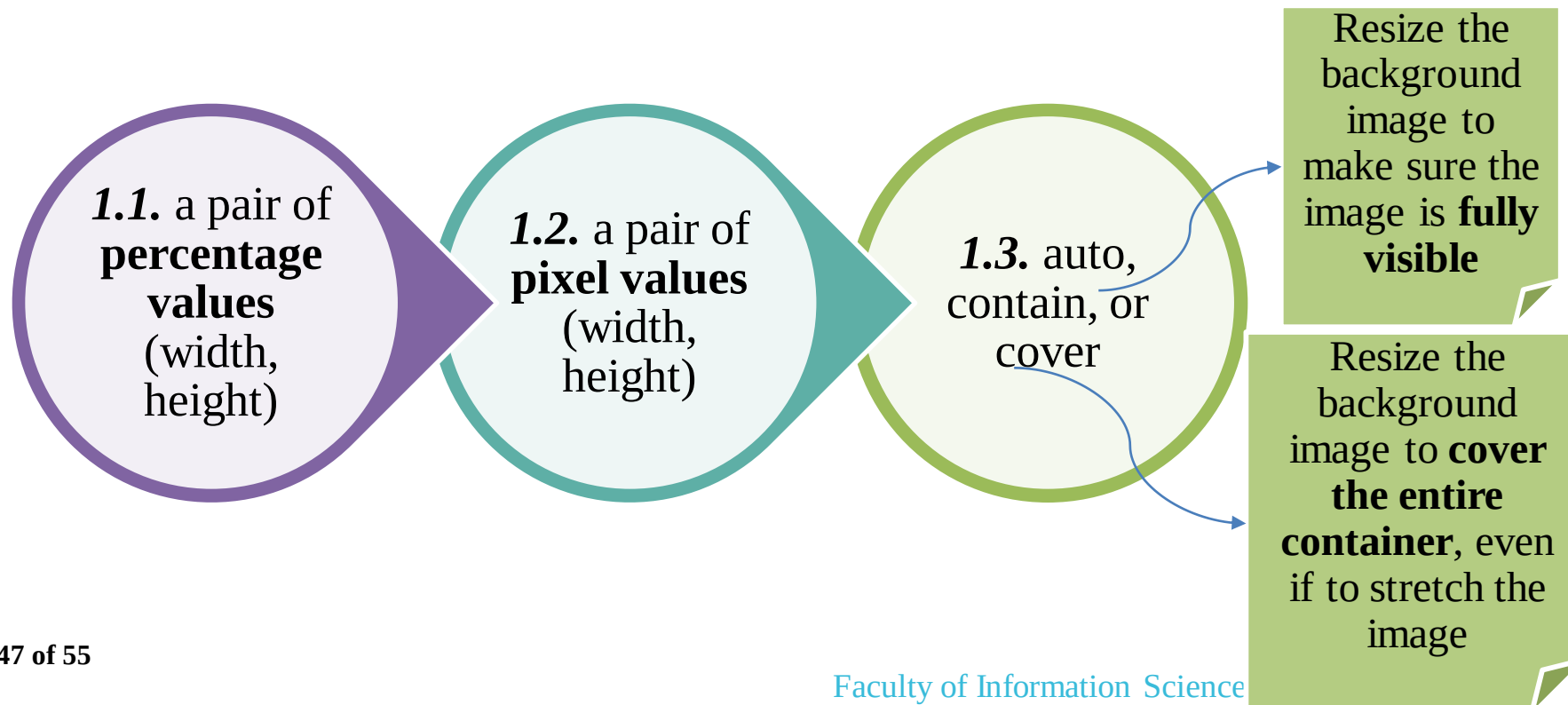


CSS3 Background Properties

Property	Description	Example
background	A shorthand property for setting all the background properties in one declaration	background: yellow; background: url(img_flwr.gif);
background-clip	Specifies the painting area of the background	background-clip: content-box;
background-image	Specifies one or more background images for an element	background-image: url(img_flwr.gif), url(paper.gif);
background-origin	Specifies where the background image(s) is/are positioned	background-origin: content-box;
background-size	Specifies the size of the background image(s)	background-size: contain;

CSS3 background-size Property

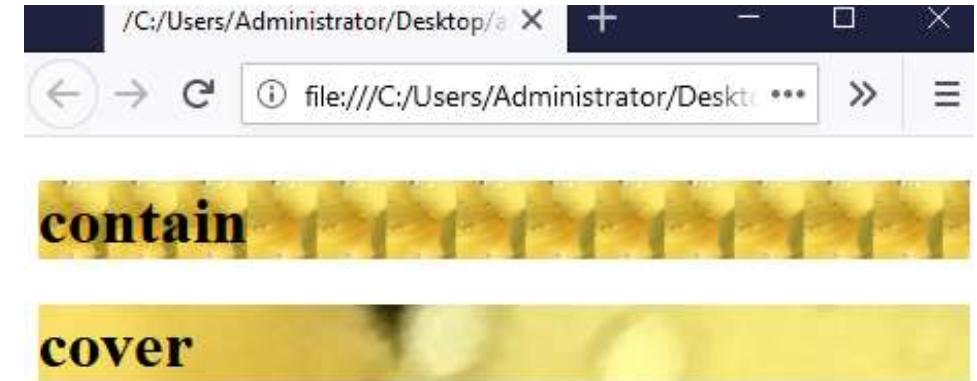
- The CSS3 **background-size** property allows you to specify the **size** of background images.





background-size Example

```
Untitled - Notepad
File Edit Format View Help
<!DOCTYPE html>
<html>
  <head>
    <style>
      #div1 { background: url(image.jpg);
              background-size: contain;
              background-repeat: repeat;}
      #div2 { background: url(image.jpg);
              background-size: cover;
              background-repeat: repeat;}
    </style>
  </head>
  <body>
    <h1 id= "div1"> contain</h1>
    <h1 id= "div2"> cover</h1>
  </body>
</html>
```





CSS3 Multiple Background Images

■ CSS3 allows you to add multiple background images for an element, through the **background-image** property.

■ The different background images are separated by **commas**, and the images are stacked on top of each other, where the first image is closest to the viewer.



Multiple Background Example

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
  <head>
    <style>
      #div1 { border: 2px solid black;
        padding: 50px;
        background: url(bulb.png), url(image.jpg);
        background-repeat: no-repeat;
        background-size: contain, cover;}
    </style>
  </head>
  <body>
    <h1 id= div1> Here we have two background
      images.</h1>
  </body>
</html>
```





CSS3 Rounded Corners Properties

Property	Description
border-radius	A shorthand property for allowing to add rounded corners to elements!
border-top-left-radius	Defines the shape of the border of the top-left corner
border-top-right-radius	Defines the shape of the border of the top-right corner
border-bottom-right-radius	Defines the shape of the border of the bottom-right corner
border-bottom-left-radius	Defines the shape of the border of the bottom-left corner



The CSS3 opacity Property

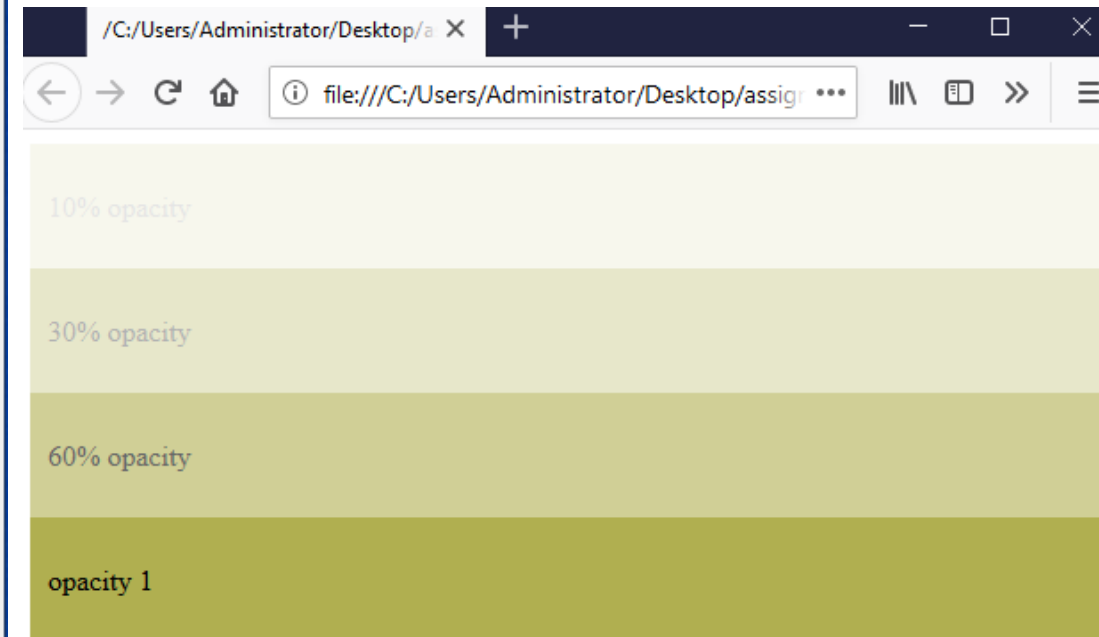
- The opacity property sets the opacity level for an element.
- The opacity-level **describes the transparency-level**, where
 - ✓ 1 is not transparent at all,
 - ✓ 0.5 is 50% see-through, and
 - ✓ 0 is completely transparent.



opacity Example

```
Untitled - Notepad
File Edit Format View Help

<!DOCTYPE html>
<html>
<head> <style>
div {
    background: rgb(176, 175, 80); padding: 10px; }
</style> </head>
<body>
<div style="opacity:0.1;"><p>10% opacity</p></div>
<div style="opacity:0.3;"><p>30% opacity</p></div>
<div style="opacity:0.6;"><p>60% opacity</p></div>
<div><p>opacity 1</p></div>
</body>
</html>
```





Lets practice!

Write HTML code for the following CSS code.

1. `#example1 { border-radius: 25px; background: #8AC007; padding: 20px; width: 200px; height: 150px; }`
2. `#example2 { border-radius: 25px; border: 2px solid #8AC007; padding: 20px; width: 200px; height: 150px; }`
3. `#example3 { border-radius: 25px; background: url(paper.gif); background-position: left top; background-repeat: repeat; padding: 20px; width: 200px; height: 150px; }`



Summary

Lecture 3.3: Visual Element and Graphics

- ✓ background-clip Property
- ✓ background-origin Property
- ✓ background Properties
- ✓ background-size Property
- ✓ Multiple Background Images
- ✓ Rounded Corners Property
- ✓ Opacity Property

